GRADE 8

STEM OLYMPIAD

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

Sample paper

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is (-3/5) ÷ (9/10)?

A. -2/3
B. -3/2
C. 2/3
D. 3/2

Q2. Solve: 5x - 9 = 3x + 17.

A. x=11
B. x=12
C. x=13
D. x=14

Q3. What is sqrt(2025)?

A. 35
B. 40
C. 45
D. 50

Q4. What is the cube root of 3375?

A. 10
B. 12
C. 15
D. 25

Q5. Simplify: 2^5 x 4^3 ÷ 2^4.

A. 2^5
B. 2^6
C. 2^7
D. 2^8

Q6. A price increases from Rs 800 to Rs 920. Percentage increase is:

A. 12%
B. 15%
C. 18%
D. 20%

Q7. A cylinder has radius 3 cm and height 10 cm. Using pi=3.14, its volume is about:

A. 94.2 cm^3
B. 188.4 cm^3
C. 282.6 cm^3
D. 314 cm^3

Q8. A rectangle is enlarged so length and width both become 1.5 times. Its area becomes:

A. 1.5 times
B. 2 times
C. 2.25 times
D. 3 times

Q9. The mean of 5 numbers is 18. If four are 12, 16, 20 and 22, the fifth is:

A. 18
B. 20
C. 22
D. 24

Q10. Probability of drawing an ace from a standard 52-card deck is:

A. 1/52
B. 1/26
C. 1/13
D. 4/13

Q11. If 8 pumps fill a tank in 6 h, 12 identical pumps fill it in:

A. 3 h
B. 4 h
C. 6 h
D. 9 h

Q12. The diagonal of a rectangle divides it into two triangles of:

A. equal area
B. always different area
C. zero area
D. unrelated area

Q13. A number is written as 5.6 x 10^4. Its ordinary form is:

A. 5,600
B. 56,000
C. 560,000
D. 0.00056

Q14. Which expression is a perfect square?

A. 2^3 x 3^2
B. 2^4 x 3^2
C. 2^5 x 3
D. 2 x 3^3

Q15. A graph rises 12 units when x rises 3 units. Its constant rate of change is:

A. 3
B. 4
C. 9
D. 36

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Microorganisms that convert milk into curd are mainly:

A. Lactobacillus bacteria
B. viruses
C. algae only
D. protozoa only

Q17. Why is water generally used to extinguish a fire?

A. It increase the temperature
B. It lowers the temperature of the burning substance
C. It Supplies Oxygen
D. It acts as a fuel

SampleQPaper

Q18.A Farmer wants to store grains for several?

A. Adding water to grains
B. Drying the grains exposed properly before storage
C. Leaving grains exposed to rain
D. Keeping wet grains in a closed container

Q19.Suppose the world continues to consume fossil fuels at the present rate without developing alternatives. Which problems is mot likely to become serious in the long term?

A. Fossil fuel reserve may become depleted
B. Oxygen will completely disappears
C. More fossil fuel will form quickly
D. Petroleum will become renewable

Q20. Friction can be reduced in machine parts by:
A. lubrication
B. roughening surfaces
C. adding sand
D. increasing load

Q21. A School installs curtain s and cerpets in the its auditorium to reduce unwanted sound . How do these materials help?
A.They increase sound reflection
B. They absorb some sound and reduce reflections
C. They increase the frequency of sound
D. They produce louder sound

Q22. The basic structural and functional unit of life is the:
A. cell
B. tissue
C. organ
D. organ system

Q23. In humans, fertilisation usually occurs in the:
A. oviduct
B. stomach
C. kidney
D. large intestine

Q23. A lightning conductor protects a building by providing:
A. a safe conducting path to Earth
B. more friction
C. a source of lightning
D. an insulating cloud

Q24. A force of 50 N acts on area 0.5 m^2. Pressure is:

A. 25 Pa
B. 50 Pa
C. 100 Pa
D. 250 Pa

Q25. **Two students are caught outside during a thunderstorm student. A stand under a large tree, while student B moves away from the tree and crouches low in a safer location until shelter is available. Which action is safer?**
A. Student A, because trees provide protection
B.Student B, because tall is dated objects can be struck by lighting
C. Both are equally safe
D.Neither student is affected by lightning

Q26. A Student shines a torch on a plane mirror at an angle of 30° with the normal. At what angle will the reflected ray make with the normal?
A. 15°
B. 30°.
C. 60°
D. 90°

Q27. A jeweler wants to give a silver appearance to a cheaper metal ornament. Which process can be used?
A. Filtration
B. Electroplating
C. Evaporation
D. Sedimentation

Q28. Increasing greenhouse gases tends to:
A. increase global warming
B. stop all sunlight
C. freeze all oceans immediately
D. remove the atmosphere

Q29. A farmer removes trees from a will side to create agricultural land. During heavy rain the top soil is washed away. What is the main reason?
A. Trees increase soil erosion
B. Trees roots keep hold the soil together
C. Trees prevent water from entering soil
D. Soil becomes heavier after deforestation.

Q30. Crop rotation helps maintain soil fertility by:
A. varying nutrient demands and including useful crops such as legumes
B. removing all microbes
C. using the same crop forever
D. preventing all water entry

Sample Paper

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A hydraulic lift works mainly because pressure in an enclosed fluid is:

A. transmitted throughout the fluid
B. lost completely
C. converted only to heat
D. independent of force

Q32. A 12 V supply across a 6 ohm resistor gives current:

A. 0.5 A
B. 2 A
C. 6 A
D. 72 A

Q33. A machine receives 800 J and gives 600 J useful output.

Q36. Which graph best compares categories such as crop yield of four fields?

A. Bar graph
B. Line graph only
C. Scatter plot only
D. Number line

Q37. A model bridge fails at its joints. Which redesign directly targets the weakness?

A. Reinforce the joints
B. Make spans longer
C. Add weight at the centre
D. Remove cross-bracing

Efficiency is:

A. 60%
B. 70%
C. 75%
D. 80%

Q34. A vehicle travels 72 km/h. This equals:

A. 10 m/s
B. 15 m/s
C. 20 m/s
D. 25 m/s

Q35. A wind turbine's blade length is doubled while shape and wind remain similar. The swept area becomes approximately:

A. 2 times
B. 3 times
C. 4 times
D. 8 times

Q38. A robot moves 8 m East, 6 m North, 3 m West, 2 m South. Its final coordinates relative to start are:

A. 5 m East, 4 m North
B. 11 m East, 8 m North
C. 5 m West, 4 m North
D. 8 m East, 2 m North

Q39. A student tests whether salt affects boiling point. The controlled variable should include:

A. same amount of water and heating setup
B. different amounts of water
C. different thermometers each time
D. different heating powers

Q40. A solar panel is rated 200 W and receives full Sun for 5 h. Ideal energy is:

A. 0.2 kWh
B. 0.5 kWh
C. 1.0 kWh
D. 5 kWh

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A 3:1 gear reduction means the output shaft typically turns:

A. slower with greater torque
B. faster with less torque always
C. at identical speed
D. without torque

Q42. Which material is best for a heat-insulating container?

A. Foam
B. Copper
C. Aluminium
D. Iron

Q43. A child accidentally gets an unknown liquid in her hand. Before touching it further, she wants to determine whether it is acidic or basic. Which method is appropriate for a school laboratory?

A. Taste the liquid
B. Smell it directly
C. Test it by using a suitable indicator
D. Touch it with bare hands

Q44. A 2 kg cart accelerates at 3 m/s². Net force is:

A. 1.5 N
B. 5 N
C. 6 N
D. 9 N

Q45. A dataset is 40, 42, 39, 41, 120. Which value is an outlier?

A. 39
B. 40
C. 42
D. 120

Q46. Which design process is most appropriate?

A. Identify need - brainstorm - prototype - test - improve
B. Build once - never test
C. Copy blindly - decorate
D. Test before identifying the problem

Q47. A convex lens can form a real image when the object is:

A. beyond the focal point
B. always at the lens centre only
C. behind an opaque screen
D. inside the mirror

Q48. An object floats in water when the upward buoyant force is:

A. equal to its weight in equilibrium
B. always zero
C. less than its weight while stationary
D. unrelated to weight

Q49. A greenhouse model traps heat mainly because transparent walls allow much incoming sunlight but reduce:

A. loss of heat from inside
B. gravity
C. air pressure to zero
D. all visible light

Q50. If a fair die is rolled twice, probability of getting 6 both times is:

A. 1/6
B. 1/12
C. 1/36
D. 1/72

Sample paper